FACULTY OF COMPUTING

SEMESTER 2/20252026

SECP3843 - SPECIAL TOPIC IN

DATA ENGINEERING

SECTION 01

TUTORIAL 1: MICROSOFT AZURE

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Project Overview

This project presents a robust Business Intelligence (BI) reporting solution designed to analyze customer distribution, product pricing, and sales demand for a bicycle and accessories enterprise. The backend data pipeline utilizes Microsoft Azure for data processing and preparation also integrating with Microsoft Power BI to deliver interactive, data-driven dashboards. The core objective of this dashboard is to empower decision-makers by visualizing key performance metrics across a customer base of 847 individuals located in 450 cities. By evaluating a catalog of 295 total products, the solution provides immediate visibility into pricing margins, salesperson performance, and revenue distribution. The resulting visualizations reveal critical business insights, such as the fact that while high-ticket items like Touring Bikes drive the majority of the revenue , everyday accessories and apparel drive the highest overall order volume.

Description on the Solution

The BI solution relies on Microsoft Azure for upstream data storage and transformation, which is then connected to Microsoft Power BI to render three primary analytical modules. Together, these modules provide a holistic view of the company's sales and operational data:

- **Customer Base and Geography**: This module maps out the global footprint of the business, highlighting a customer presence across regions including North America and Europe. It effectively tracks 847 total customers spread across 450 distinct cities. Additionally, it features a performance tracking mechanism that displays the number of customers assigned to individual salespeople.
- **Product Catalog and Pricing Strategy**: This view is designed to analyze the profitability of the company's 295 products. It provides a macro-level financial baseline by tracking the average list price at 744.60 against an average cost of 438.22. A scatter plot visualization compares the sum of cost versus the sum of price , allowing stakeholders to easily assess profit margins across various categories like Road Bikes, Mountain Frames, Touring Bikes, and Jerseys.
- **Product Demand and Volume**: Focusing on sales execution, this segment breaks down the 2K total units sold. It provides a crucial contrast between gross revenue and order quantity. The dashboard highlights that Touring Bikes and Road Bikes generate the highest gross revenue at 708.69K and 170.83K respectively. Conversely, the highest volume of individual orders comes from smaller accessories and clothing items, specifically the "Classic Vest, S", short-sleeve apparel, and dissolving bike wash.

Technology Use

Technology	Description
SQL Server Management (SSMS)	An external tool used to write SQL queries, manage database security, and verify the data inside Synapse
Resource Group	The container. A logical grouping that holds all these services together for easier management, billing, and security.
Azure Data Factory	It automates the movement and scheduling of data between sources and processing tools.
Azure Storage Account (Azure Data Lakes Gen2)	The data lake. This provides the physical storage for raw, semi-processed, and final data files.
Azure Databricks	The transformation engine. An Apache Spark-based platform used for heavy data cleaning, engineering, and advanced analytics.
Azure Synapse Analytics	The data warehouse. It serves as the final destination where processed data is stored for high-speed SQL querying and BI reporting.
Azure Key Vault	It securely stores secrets like database passwords and connection strings so they aren't hardcoded in scripts.
Power BI	Connects to Synapse to create the final interactive dashboards and visual reports for stakeholders.

Table 1 Name of Technology Used with Its Description

Table 1 shows that all the technologies used to complete Azure end-to-end data lifecycle from managing the data to visualizing it for business decisions. This table also follows a Modern Data Warehouse (MDW) architecture which is a standard framework for high-scale data engineering.

Data Pipeline Architecture

The data pipeline for this project is built upon a Modern Data Warehouse (MDW) framework and utilizes the Medallion Architecture pattern to systematically ingest, transform, and serve data. The workflow is divided into several distinct stages:

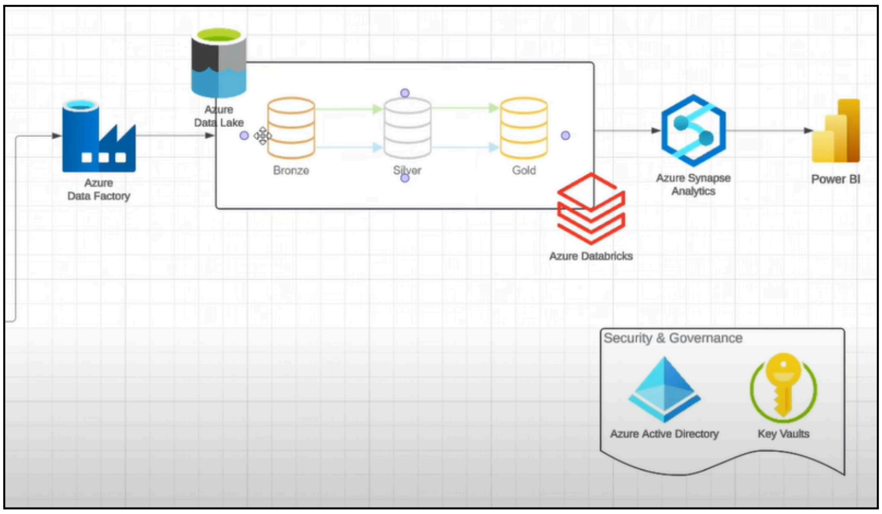

Figure 1 Data Pipeline Architecture

1. Data Ingestion (Azure Data Factory & Azure Data Lake):

The pipeline begins with Azure Data Factory, which acts as the orchestration engine. It securely extracts raw data from the source systems and ingests it into the Bronze layer of the Azure Data Lake.

2. Data Transformation (Azure Databricks):

The core data processing is handled by Azure Databricks, an Apache Spark-based analytics platform. It processes the data through progressive validation layers:

- Bronze to Silver: Raw data from the Bronze layer is cleansed, filtered, and standardized, then stored in the Silver layer.
- Silver to Gold: The cleansed Silver data undergoes heavy business-level transformations and aggregations. The highly refined, business-ready data is then saved in the Gold layer.

3. Data Serving (Azure Synapse Analytics):

   The refined data from the Gold layer is connected to Azure Synapse Analytics. Synapse acts as the enterprise data warehouse, providing a highly scalable and optimized environment for executing fast, complex SQL queries.

4. Data Visualization (Power BI):

   Power BI connects directly to Azure Synapse Analytics to render the final interactive dashboards and visual reports. This allows stakeholders to interact with the data and extract actionable business insights.

5. Security & Governance:

   Overarching the entire pipeline is a robust security layer. Azure Active Directory is used to manage identities, user access, and permissions across all services. Concurrently, Azure Key Vault is utilized to securely store and manage sensitive information, such as database credentials, secrets, and connection strings, ensuring they are never hardcoded into the pipeline scripts.

MS Power BI Dashboard

The database named AdventureWorks2025 was used to do a dashboard after going through all the 3-layer of medallion architecture which means this dataset is already transformed from raw, unverified data into highly refined, structured, and business-ready data optimized for reporting, analytics and machine learning.

1) **Customer Base and Geography**

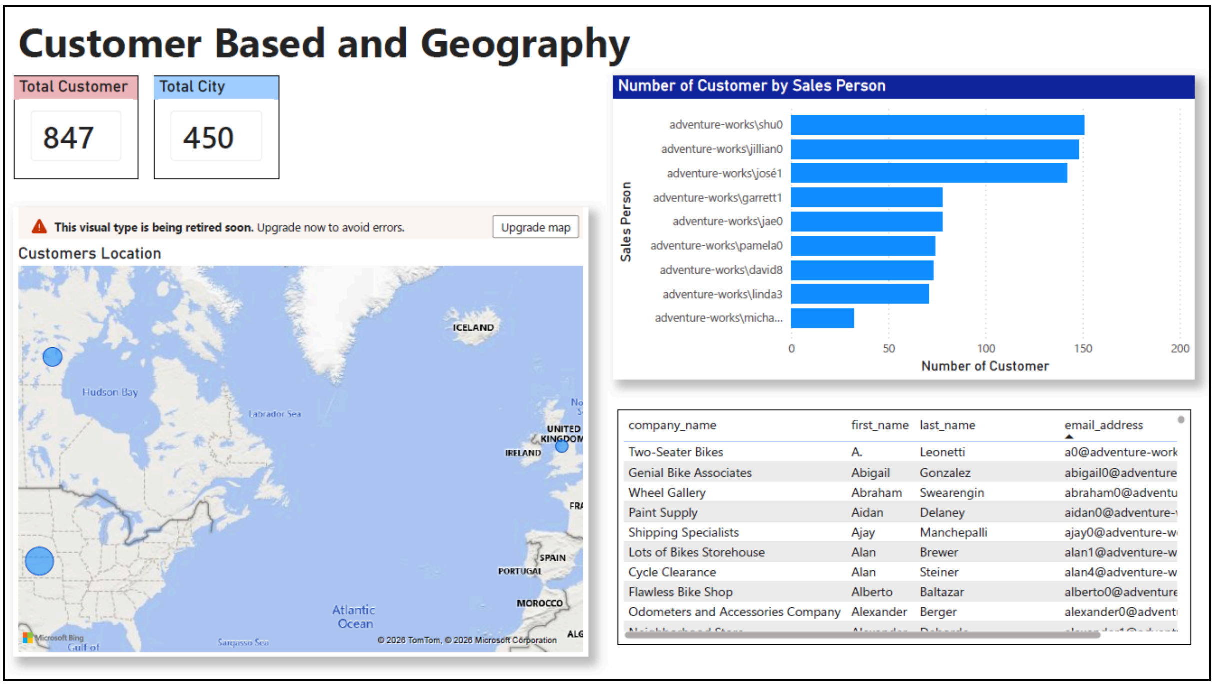

Figure 2 Customer Base and Geography Dashboard

Figure 2 shows the dashboard presents a clear, high level narrative of customer distribution and sales ownership, enabling quick insights into both geographic reach and sales performance. At a glance, the key KPIs highlight the total number of customers and cities covered, establishing the scale of operations. The geographic visualization then answers the question of "Where are our customers located?" by mapping customer presence across regions, making it easy to identify concentration areas and potential gaps. Complementing this, the sales performance chart addresses "Who is managing them?" by showing the number of customers handled by each salesperson, allowing for a direct comparison of workload and performance. Finally, the detailed data grid provides granular, record level information to support deeper analysis and validation. Together, these elements create a cohesive story that connects overall business reach with individual sales contributions, supporting more informed decision-making.

2) **Product Catalog and Pricing Strategy**

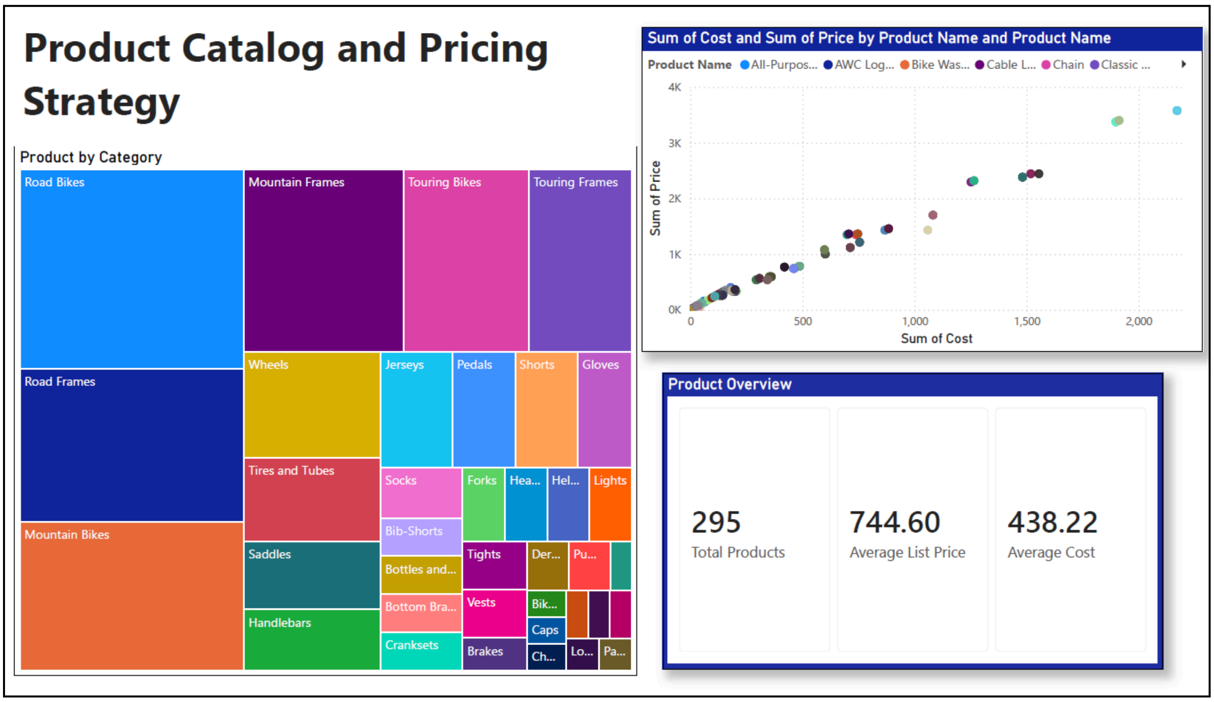

Figure 3 Product Catalog and Pricing Strategy Dashboard

Figure 3 shows a dashboard that tells a cohesive story of how the product portfolio is structured and how pricing decisions impact profitability. It begins with a high level product overview, where key metrics such as total products, average list price, and average cost establish a baseline understanding of the catalog's scale and overall margin potential. From there, the treemap visualization breaks down the portfolio by category, revealing which product groups dominate in terms of volume or value and highlighting the company's strategic focus areas. The narrative then deepens with the cost-versus-price scatter plot, which provides a more analytical perspective on pricing strategy by showing how individual products are positioned in relation to their costs. This allows patterns, outliers, and margin consistency to be easily identified. Together, these elements connect the structure of the product catalog with pricing behavior, offering a clear view of how product mix and pricing strategy work together to drive profitability.

3) **Product Demand and Volume**

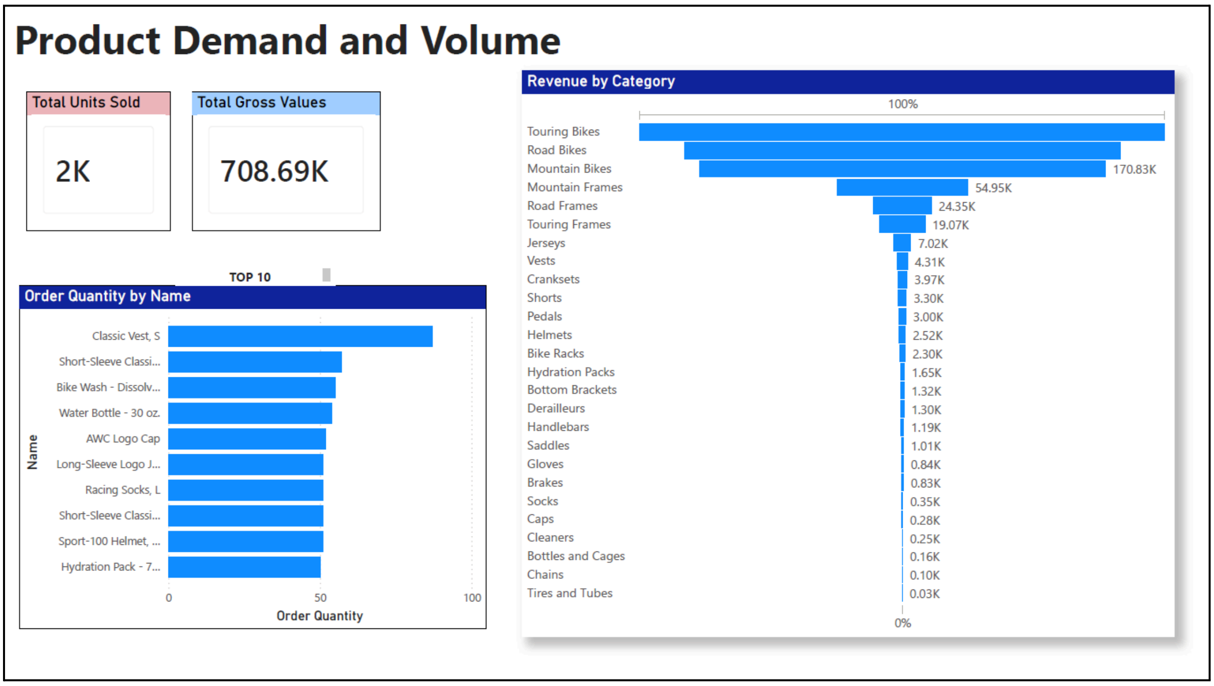

Figure 4 Product Demand and Volume Dashboard

Figure 4 presents the “Product Demand and Volume” dashboard, which clearly illustrates the relationship between revenue generation and sales volume within AdventureWorks. The overall performance indicators show that approximately 2K units have been sold, contributing to a total gross value of 708.69K, suggesting a strong revenue outcome relative to the sales volume. A deeper look into the “Revenue by Category” visualization reveals that high-value products such as bikes such as Touring Bikes, Road Bikes, and Mountain Bikes which dominate revenue contribution. These categories generate significantly higher income despite not necessarily being sold in the largest quantities.

In contrast, the “Order Quantity by Name” chart highlights a different pattern, where lower-priced items such as vests, jerseys, water bottles, and caps appear among the top 10 most frequently purchased products. This indicates that while these accessories do not contribute as much individually to total revenue, they experience higher demand and drive overall sales volume. Together, these insights demonstrate that typical premium items are the primary drivers of revenue, affordable and everyday accessories sustain consistent purchasing activity and customer engagement.

Reflection

1. Muhammad Afiq Danial Bin Rozaidie

From this tutorial I learned how to build an end-to-end data pipeline on Microsoft Azure from scratch with my teammates. Building this data pipeline in Azure taught me the real challenge of being a data engineer while it is not just about connecting services, but developing a seamless flow from raw ingestion to Power BI insights. I hope this whole experience will significantly sharpen my knowledge to design end-to-end architectures and bring me confidence to face a real challenge in the real world.

2. Muhammad Mukhritz Al Iman Bin Mohd Raffi

Through this project, I gained profound hands-on experience in implementing a Modern Data Warehouse using the Medallion architecture on Azure. One of the main challenges we faced was configuring the data transformations within Azure Databricks and ensuring smooth, accurate data transitions between the Bronze, Silver, and Gold layers. By collaborating closely with my teammates and debugging our workflows step-by-step, we successfully resolved these pipeline bottlenecks. This teamwork not only improved our collective problem-solving skills but also deepened my technical understanding of how data engineering services orchestrate securely and efficiently.

3. Ahmad Adib Zikri bin A.Mazlam

This tutorial of Microsoft Azure shows me more to understand on how the data pipeline can be built. I always know but now I realized that it is not as simple as that to be a data analyst. Imagine from data transmissions to data transitions need to be accurate and smooth. Yes, that is the challenge that me and my groupmate faced. So, I hope that the excellent work from each of us will develop more of our real life problem solving skills. For me, I hope it will enhance the skills that need to for me to be a master on Microsoft Azure and MS Power BI.